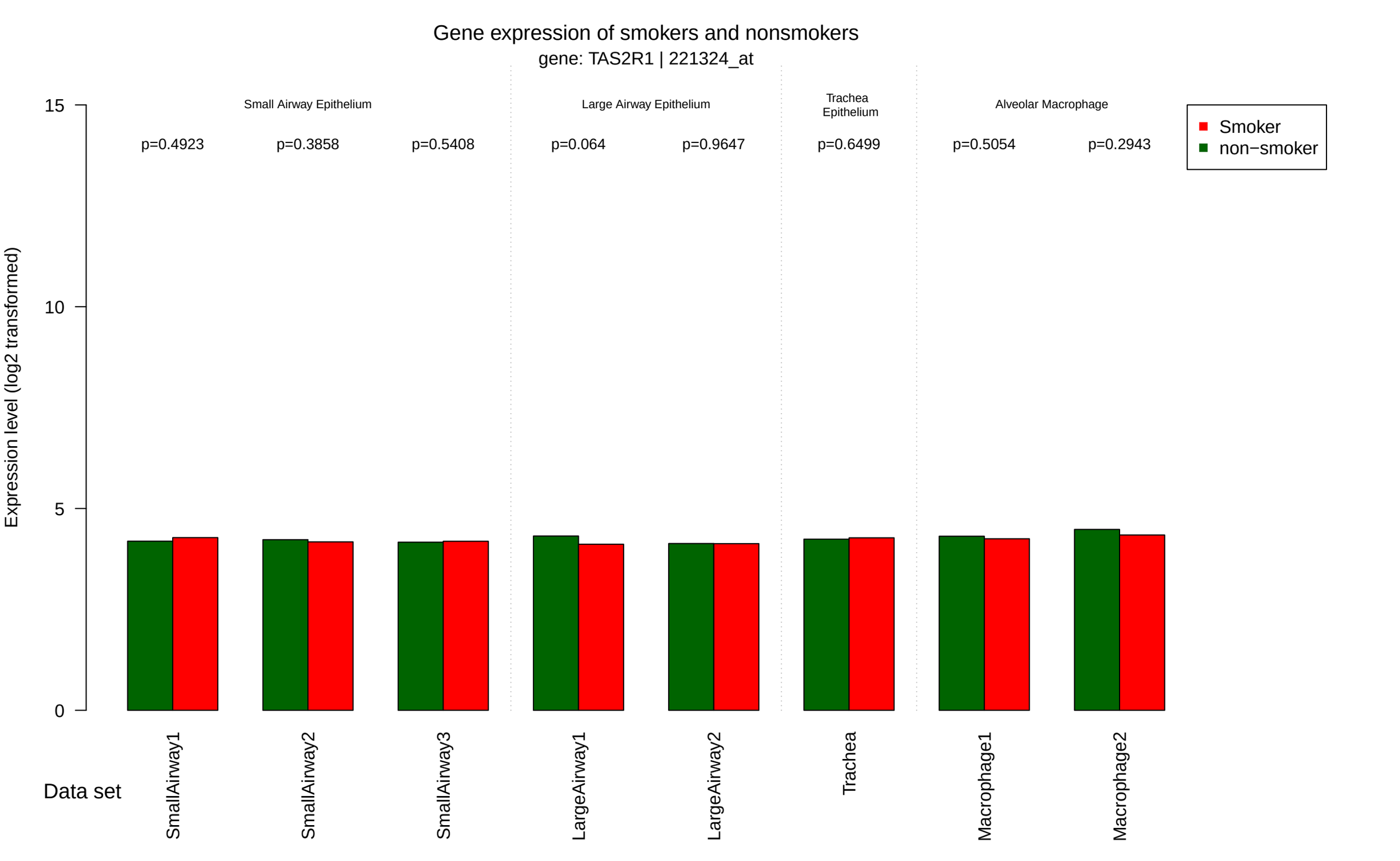
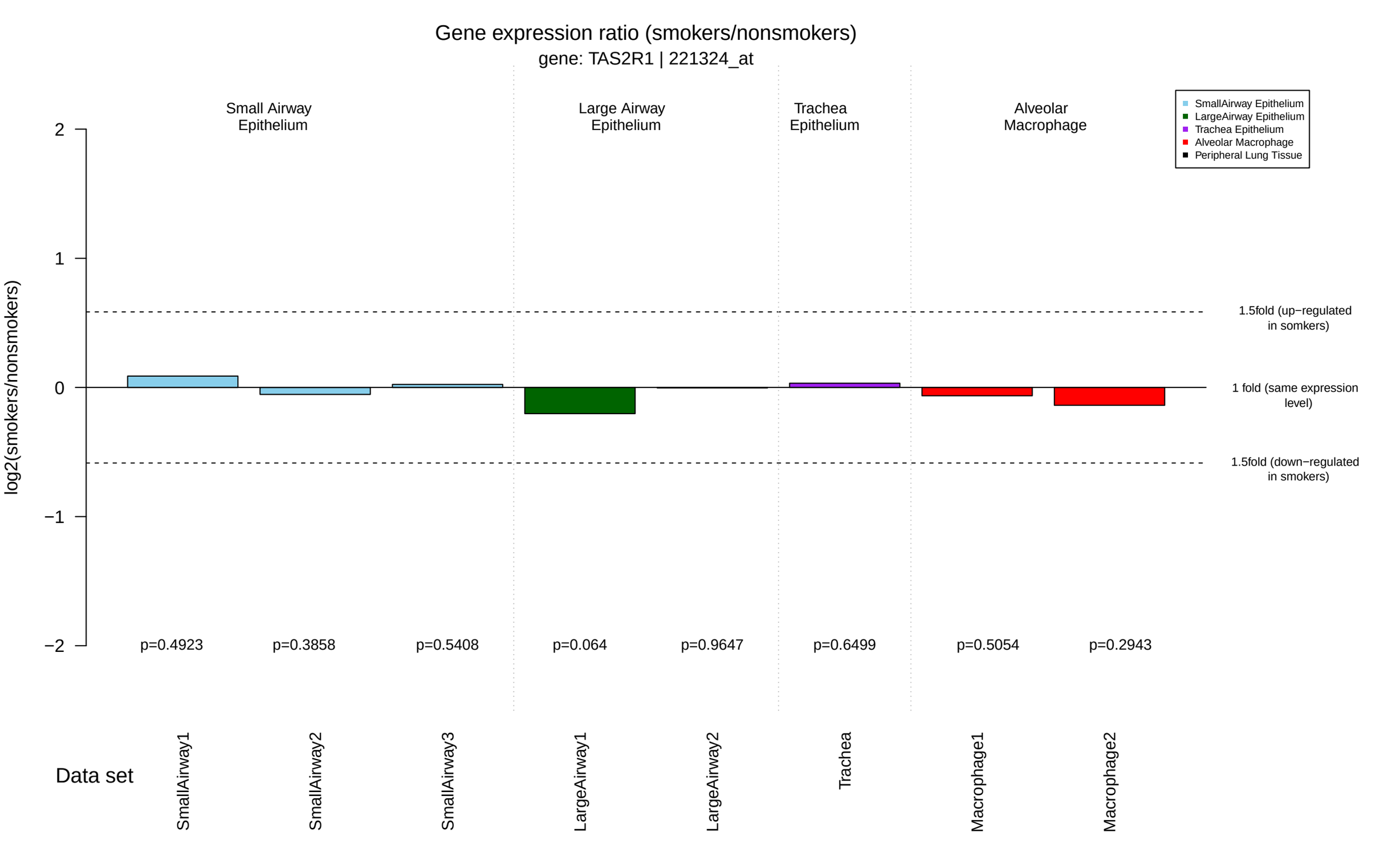
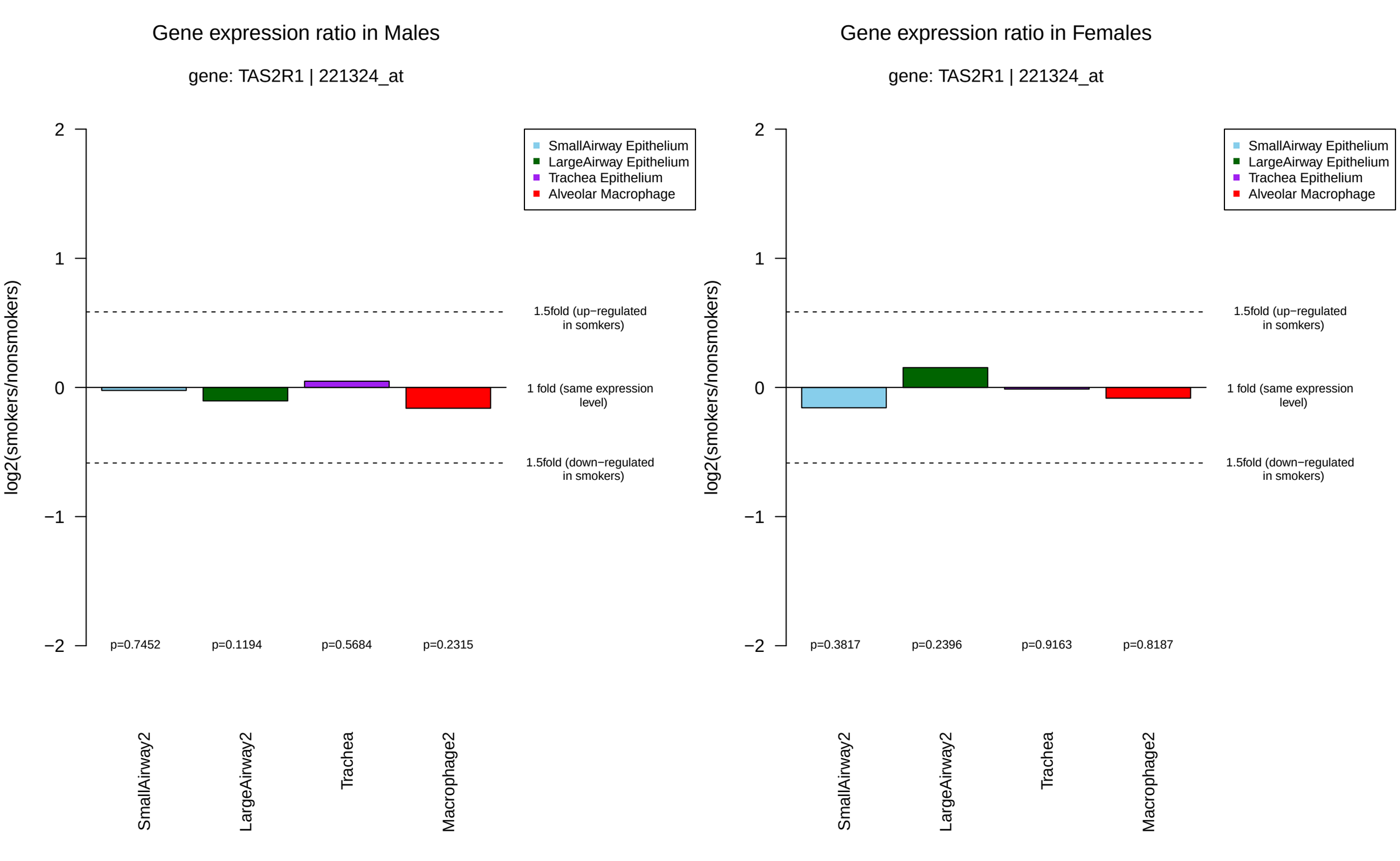
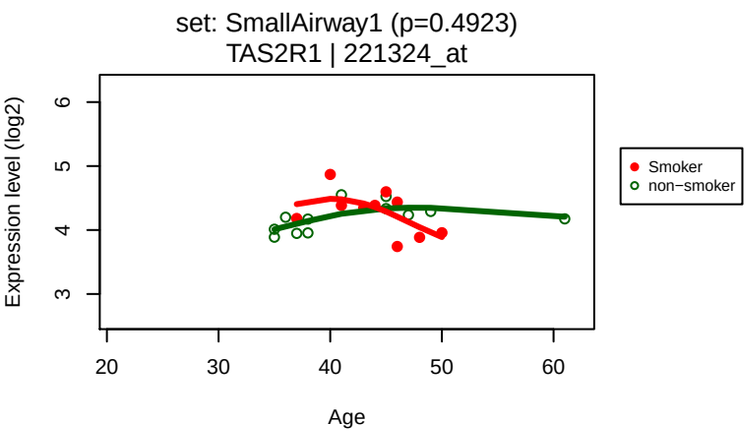
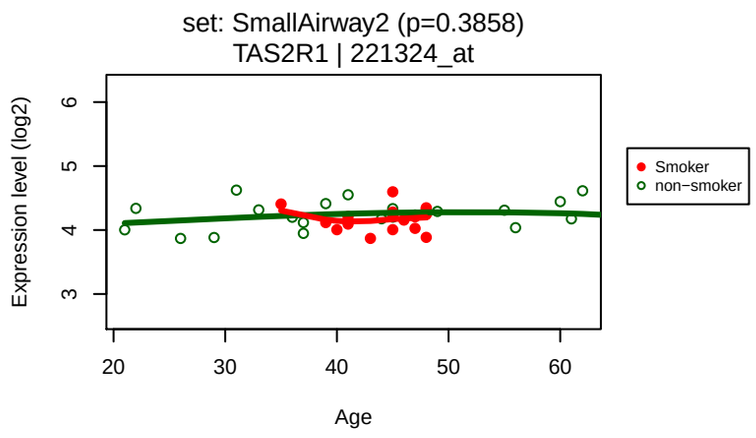
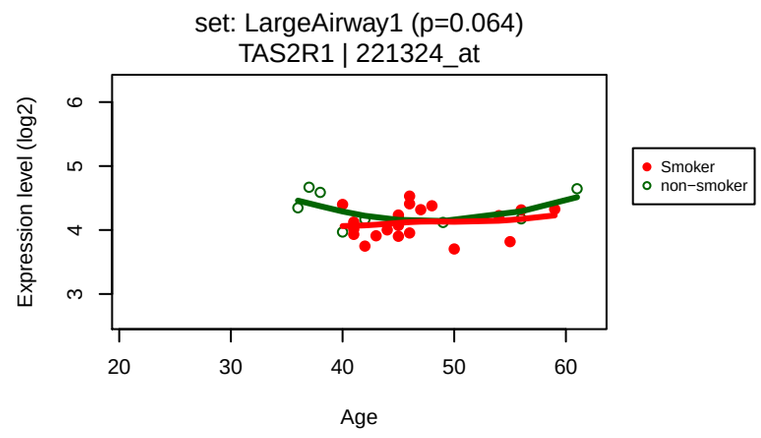

Gene expression of smokers and nonsmokers
gene: TAS2R1 | 221324_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4923
p=0.3858
p=0.5408
p=0.064
p=0.9647
p=0.6499
p=0.5054
p=0.2943
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

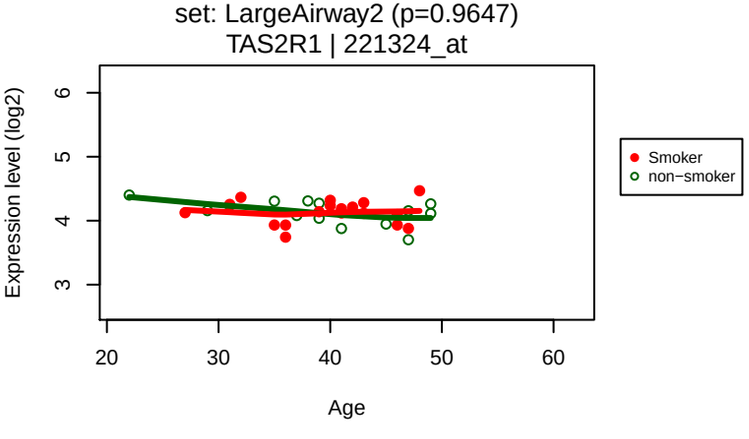

Gene expression ratio (smokers/nonsmokers)
gene: TAS2R1 | 221324_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4923
p=0.3858
p=0.5408
p=0.064
p=0.9647
p=0.6499
p=0.5054
p=0.2943
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

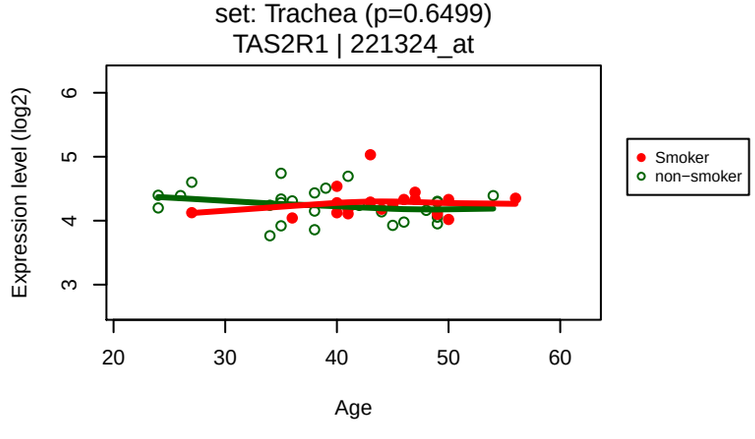

Gene expression ratio in Males
gene: TAS2R1 | 221324_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7452
p=0.1194
p=0.5684
p=0.2315
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAS2R1 | 221324_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3817
p=0.2396
p=0.9163
p=0.8187
SmallAirway2
LargeAirway2
Trachea
Macrophage2

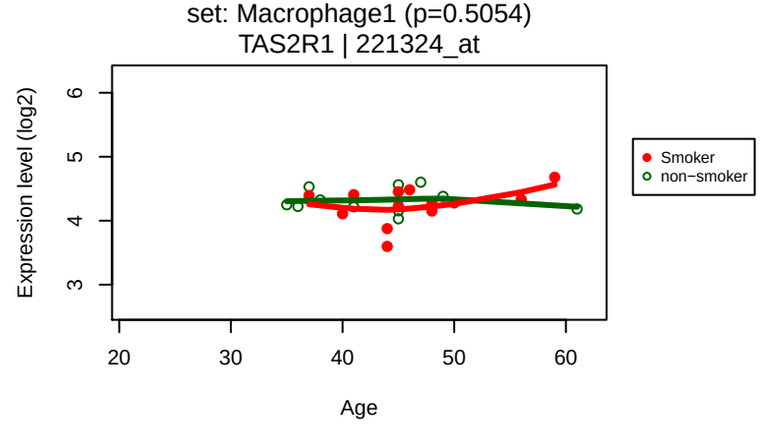

set: SmallAirway1 (p=0.4923)
TAS2R1 | 221324_at
Expression level (log2)
Age
Smoker
non−smoker

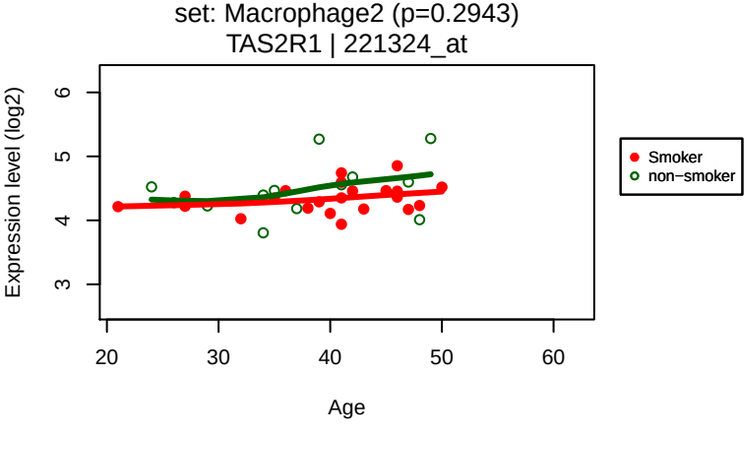

set: SmallAirway2 (p=0.3858)
TAS2R1 | 221324_at
Expression level (log2)
Age
Smoker
non−smoker

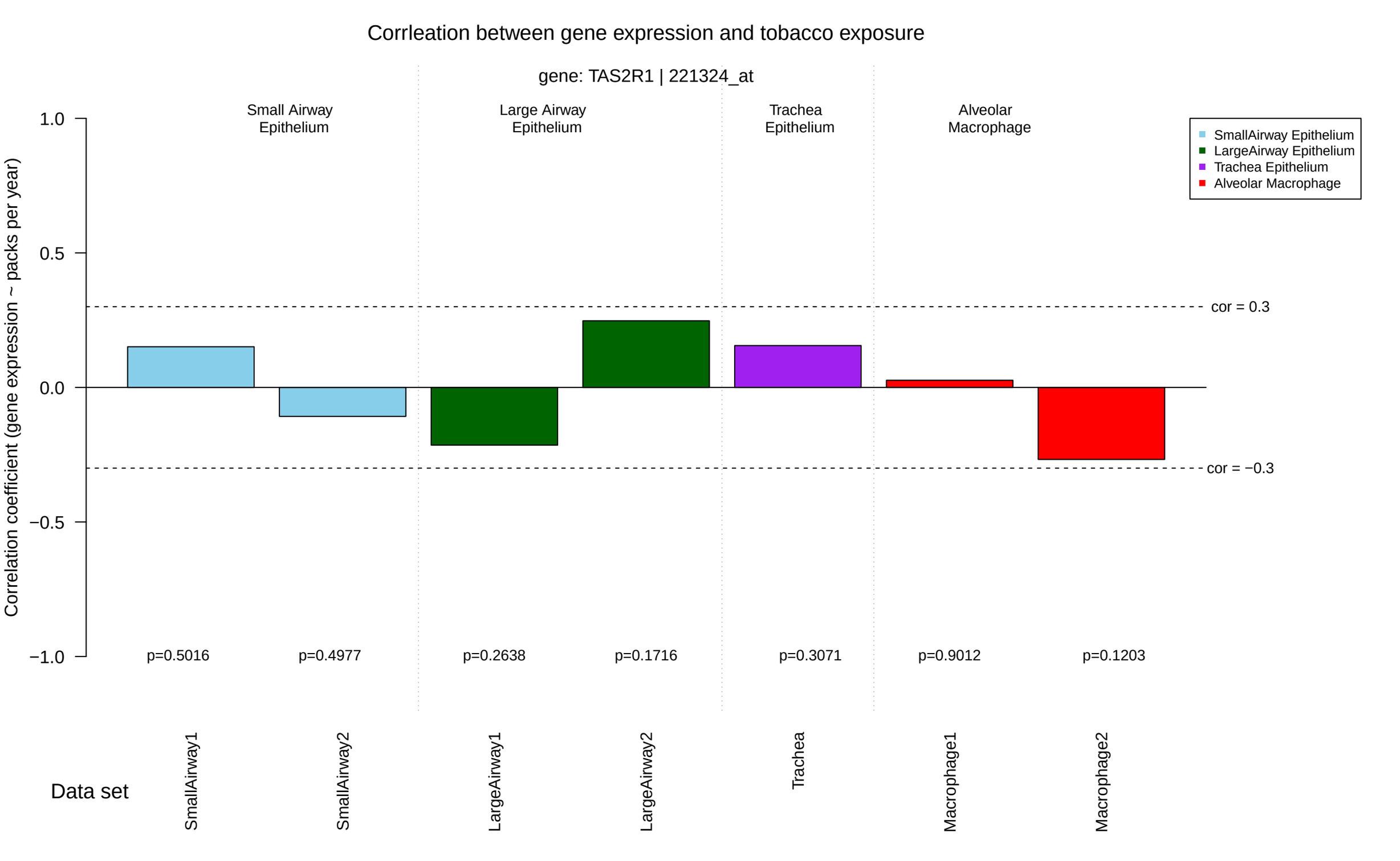

set: LargeAirway1 (p=0.064)
TAS2R1 | 221324_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9647)
TAS2R1 | 221324_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6499)
TAS2R1 | 221324_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5054)
TAS2R1 | 221324_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2943)
TAS2R1 | 221324_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAS2R1 | 221324_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5016
p=0.4977
p=0.2638
p=0.1716
p=0.3071
p=0.9012
p=0.1203
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2